GEOM90007 Assignment 2 - Design Summary
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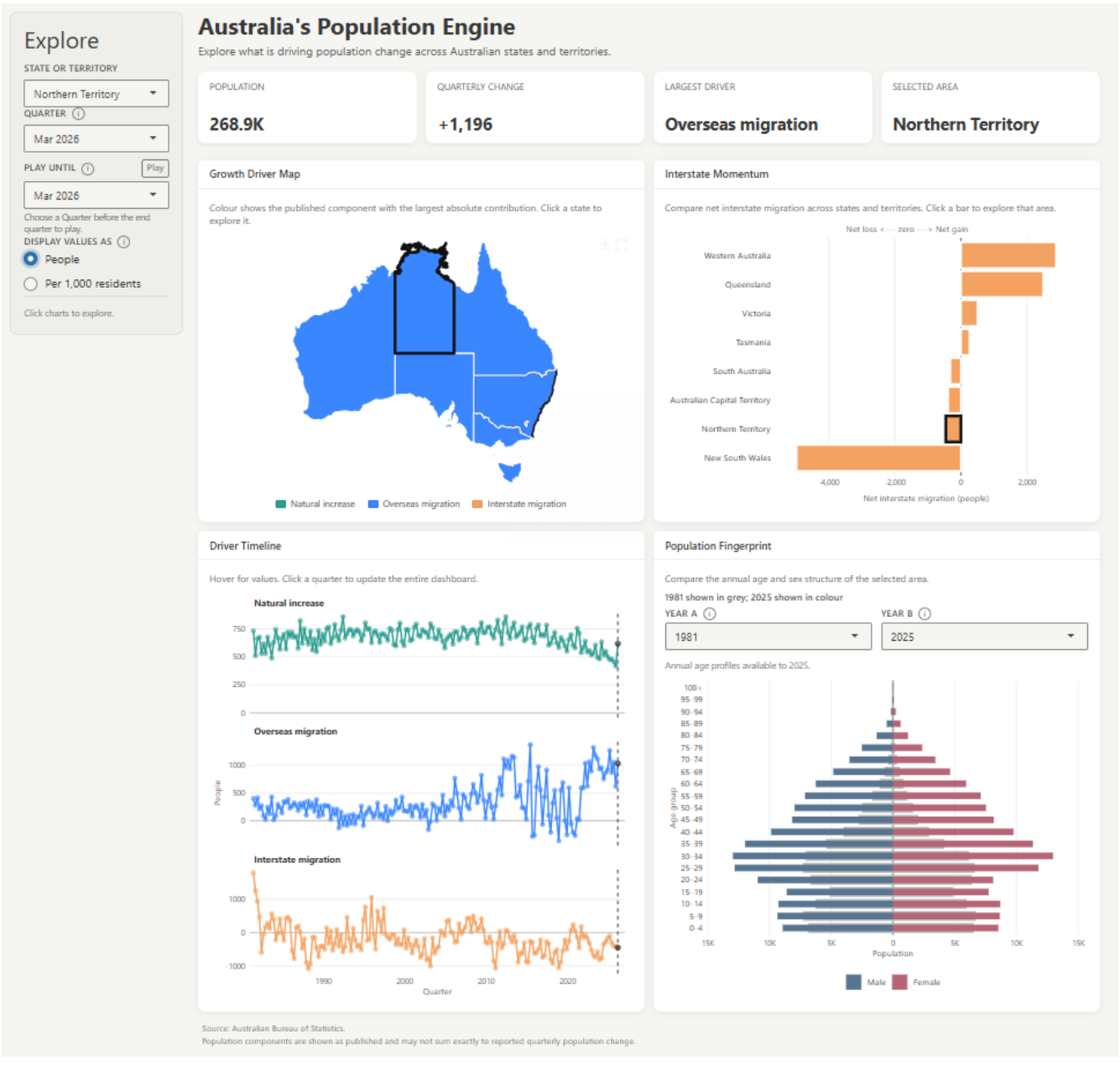
PURPOSE & AUDIENCE

Australia's Population Engine is an exploratory interface designed for a general Australian audience interested in understanding what is driving population change across states and territories. Instead of presenting population growth as a single figure, the dashboard separates natural increase, overseas migration and interstate migration so users can explore how these contributors vary across place and time. The interface also connects these patterns with interstate movement and changes in age and sex structure.

DATA & TRANSFORMATION

The interface uses Australian Bureau of Statistics population data, including quarterly population-change components, quarterly population totals and annual age-sex profiles. Single-year ages are grouped into five-year bands to make demographic structure easier to compare. A People / Per 1,000 residents control supports both absolute interpretation and fairer comparison between differently sized states. Population components are displayed as published by the ABS and may not sum exactly to the reported quarterly population change.

FINAL DASHBOARD



VISUAL DESIGN

The Growth Driver Map provides a spatial overview, with a consistent semantic palette: teal for natural increase, blue for overseas migration and orange for interstate migration. The Driver Timeline uses aligned small multiples so the three components can be compared through time without overlapping lines. Interstate Momentum uses a diverging bar chart around a common zero baseline, making gains and losses immediately distinguishable. The Population Fingerprint compares age-sex structure using a neutral grey Year A reference behind a narrower colored Year B profile.

INTERACTION & COORDINATION

The graphics operate as coordinated views rather than separate charts. Clicking a state on the map or Interstate Momentum updates the selected area across the dashboard, while clicking the timeline changes the active quarter. Custom tooltips provide details on demand. The Play interaction moves from the selected Quarter to a user-defined Play until date; smooth transitions preserve visual continuity as spatial and interstate patterns change, while the timeline remains stable and its play head moves through time. Switching between People and Per 1,000 residents supports both magnitude and population-adjusted comparison.

DESIGN PRINCIPLES & ACCESSIBILITY

The interface uses one typography system, restrained card styling and a limited color palette to maintain hierarchy and consistency. Controls are grouped in a persistent Explore panel so filtering remains separate from the analysis area. Information icons explain fewer familiar controls without adding permanent instructional text, and selected states are emphasized with outlines rather than another color encoding. Keyboard-accessible help controls and reduced-motion behavior were also included to improve accessibility.

Sources & Appendix

1. Sources used in the interface and how each source contributes to the final visualization.
2. Australian Bureau of Statistics - Components of population change
   Quarterly natural increase, net overseas migration, net interstate migration and reported population change by state and territory.
3. Australian Bureau of Statistics - Population by state and territory
   Quarterly population totals and the denominator used for values per 1,000 residents.
4. Australian Bureau of Statistics - Estimated Resident Population by age and sex
   Annual state and territory age-sex series used for the Population Fingerprint. Single ages are grouped into five-year bands for display.
5. R package: ozmaps
   Australian state and territory geometries used in the Growth Driver Map.
6. R packages: shiny, bslib, ggplot2, ggiraph, dplyr, tidyr, scales and sf
   Used to build the interface, coordinated interactive graphics, transformations, formatting and spatial visualization.

METHOD NOTE

Population components are shown as published by the ABS and may not sum exactly to the reported quarterly population change. The dashboard therefore treats the ABS 'Change Over Previous Quarter' series as the reported total and uses natural increase, net overseas migration and net interstate migration as published contributors rather than asserting an exact accounting identity.